

Vishva Arasan

925-518-6484 | varasan@ucsb.edu | linkedin.com/in/vishva-arasan-13245a279 | github.com/VishvaAr

EDUCATION

University of California, Santa Barbara

Santa Barbara, CA

B.S./M.S. Computer Science (Accelerated Track) | **GPA: 3.8**

Expected June 2028

- **Coursework:** Operating Systems, Data Structures & Algorithms, Computer Architecture, Computer Security, Automata & Formal Languages, Machine Learning

RESEARCH EXPERIENCE

Undergraduate Researcher, Software Security

Sep 2025 – Present

UCSB Verification Lab (Advisor: Prof. Tefvik Bultan)

Santa Barbara, CA

- Build STASE (Static Analysis–Guided Symbolic Execution) under the **DARPA HARDEN** program, a toolchain for detecting weird-machine / emergent-execution vulnerabilities in the **Linux kernel**.
- Designed an LLM-agent pipeline that auto-generates C test drivers for **KLEE**, achieving **10x** faster symbolic execution and **90%** fewer LLM API calls vs manual drivers.
- Developed automated bug-chaining and deep-path exploration that triggered chainable bug pairs in real-world targets including **tcpdump** and **SQLite**.

PUBLICATIONS

V. Arasan, T. Bultan. “Bug Composition: Triggering Multiple Bugs with Agentic Driver Generation.” *IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2026. *First author*.

PROFESSIONAL EXPERIENCE

Backend Software Engineer Intern

June 2026 – Sep 2026

Sesi Technologies

Remote

- Built a public **REST API** in **Python / Django REST Framework** exposing verifiable digital identity records for an agricultural network of 10,000+ users and 20,000+ mapped hectares.
- Designed an end-to-end usage metering and billing system with the **Django ORM** that tracks per-client endpoint usage, enforces rate limits, and applies configurable pricing tiers.
- Implemented **API-key authentication** and field-level response schemas that share compliance data with third parties while withholding sensitive PI.

Software Engineering Intern

Apr 2023 – Jul 2023

CUBUS Solutions Inc. (acquired by Tyfone)

Livermore, CA

- Contributed to digital banking services backed by **SQL** databases and **JavaScript** APIs.

PROJECTS

KOS Operating System Kernel | C, Kernel Threads, Semaphores

Mar 2026

- Implemented the kernel for KOS in C, including 15+ system calls for process management (**fork**, **execve**, **wait**, **exit**) and file descriptors (**dup**, **dup2**, **close**).
- Built thread-safe **IPC pipes** on circular buffers with semaphore-based synchronization, correctly blocking and waking readers and writers across processes.
- Managed process control blocks and per-process FD tables with reference counting to prevent resource leaks on **fork**, **dup**, and **exit**.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, SQL, HTML/CSS

Security & Analysis: Symbolic Execution (KLEE), Fuzzing, Static Analysis, LLM Agents for Program Analysis

Backend: Django, Django REST Framework, Flask, Node.js, PostgreSQL, SQLite, REST APIs

Tools: Linux, Git, GDB, Docker, Postman

CERTIFICATIONS & AWARDS

AWS Certified Cloud Practitioner | Amazon Web Services

Eagle Scout | Boy Scouts of America